ESE 541 Week 8 Practice Problems

Problem 1: Consider the following training and testing datasets with $p = 3$ and $N = 4$:

$$\mathcal{D}_{\text{Tr}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^4 = \left\{ \left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, 0.9 \right), \left(\begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}, 2.1 \right), \left(\begin{bmatrix} 3 \\ 9 \\ 27 \end{bmatrix}, 2.9 \right), \left(\begin{bmatrix} 4 \\ 16 \\ 64 \end{bmatrix}, 4.1 \right) \right\}$$

$$\mathcal{D}_{\text{Te}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^4 = \left\{ \left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, 1.1 \right), \left(\begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}, 1.9 \right), \left(\begin{bmatrix} 3 \\ 9 \\ 27 \end{bmatrix}, 3.1 \right), \left(\begin{bmatrix} 4 \\ 16 \\ 64 \end{bmatrix}, 3.9 \right) \right\}$$

a) Find the best linear model using best subset selection. Feel free to use software to find the parameters.

b) Find a linear model using forward stepwise selection. Feel free to use software to find the parameters.

c) Find a linear model using backward stepwise selection. Feel free to use software to find the parameters.

d) Do the models in a), b), and c) coincide?

Problem 2: Consider the following training and testing datasets with $p = 3$ and $N = 4$:

$$\mathcal{D}_{\text{Tr}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^4 = \left\{ \left(\begin{bmatrix} 1 \\ 1.1 \\ 0.9 \end{bmatrix}, 0.9 \right), \left(\begin{bmatrix} 2 \\ 1.9 \\ 2.1 \end{bmatrix}, 2.1 \right), \left(\begin{bmatrix} 3.1 \\ 3 \\ 2.9 \end{bmatrix}, 2.9 \right), \left(\begin{bmatrix} 3.9 \\ 4 \\ 4.1 \end{bmatrix}, 4.1 \right) \right\}$$

$$\mathcal{D}_{\text{Te}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^4 = \left\{ \left(\begin{bmatrix} 1 \\ 1.1 \\ 0.9 \end{bmatrix}, 1.1 \right), \left(\begin{bmatrix} 2 \\ 1.9 \\ 2.1 \end{bmatrix}, 1.9 \right), \left(\begin{bmatrix} 3.1 \\ 3 \\ 2.9 \end{bmatrix}, 3.1 \right), \left(\begin{bmatrix} 3.9 \\ 4 \\ 4.1 \end{bmatrix}, 3.9 \right) \right\}$$

a) Find the coefficients of the linear model $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$ and compute the resulting training and testing RSS.

b) Estimate the covariance matrix of X using the formula:

$$\widehat{\Sigma}_X = \frac{1}{4} \sum_{i=1}^4 (\mathbf{x}_i - \widehat{\mu}_X)(\mathbf{x}_i - \widehat{\mu}_X)^\intercal \in \mathbb{R}^{3\times 3}$$

where $\widehat{\mu}_X = \frac{1}{4}\sum_{i=1}^4 \mathbf{x}_i \in \mathbb{R}^3$.

c) Compute the largest eigenvalue and the associated eigenvector $\mathbf{v}_1$ (feel free to use computer software). Using $\mathbf{v}_1$, generate a reduced order dataset

$$\mathcal{D}_1 = \{(z_i, y_i)\}_{i=1}^4 \text{ where } z_i = \mathbf{v}_1^\intercal \mathbf{x}_i \in \mathbb{R}$$

d) Using the dataset $\mathcal{D}_1$ in c), find the coefficients of the linear model $Y = \theta_0 + \theta_1 Z$ and compute the resulting training and testing RSS.

e) Compare the training and testing RSS computed in a) with the one computed in d).

Problem 3: Which of the following statements about regularization are true? Check all that apply.

a) Using a very large value of λ cannot hurt the performance of your hypothesis. We just do not set λ to be too large to avoid computational problems.

b) Using too large a value of λ can cause your hypothesis to underfit the data.

c) Because logistic regression outputs values between 0 and 1, its range of outputs are always shrunk with an increasing λ

Problem 4: You are training a classification model with logistic regression. Which of the following statements are true? Mark all that apply.

a) Adding many new features to the model helps prevent overfitting on the training set.

b) Introducing regularization to the model always results in equal or better performance on examples not in the training set.

c) Introducing regularization to the model always results in equal or better performance on the training set.

d) Adding a new feature to the model always results in equal or better performance on the training set.